Multimodal Document Extraction System

OCR + vision LLM pipeline for invoices and receipts

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1. Objective

The goal of this project is to extract structured data from scanned invoices and receipts: invoice number, date, vendor, line items and total. The system combines OCR with a vision language model, returns schema-validated JSON, flags low-confidence fields for human review, and measures field-level accuracy on more than 30 documents.

2. System overview

Each document is preprocessed and read with Tesseract OCR. The page image, together with the OCR text as a hint, is sent to a vision LLM (Google Gemini) that returns the fields and a confidence value for each field. The output is validated with Pydantic and business rules, and a combined confidence score decides whether a field is accepted or sent to human review.

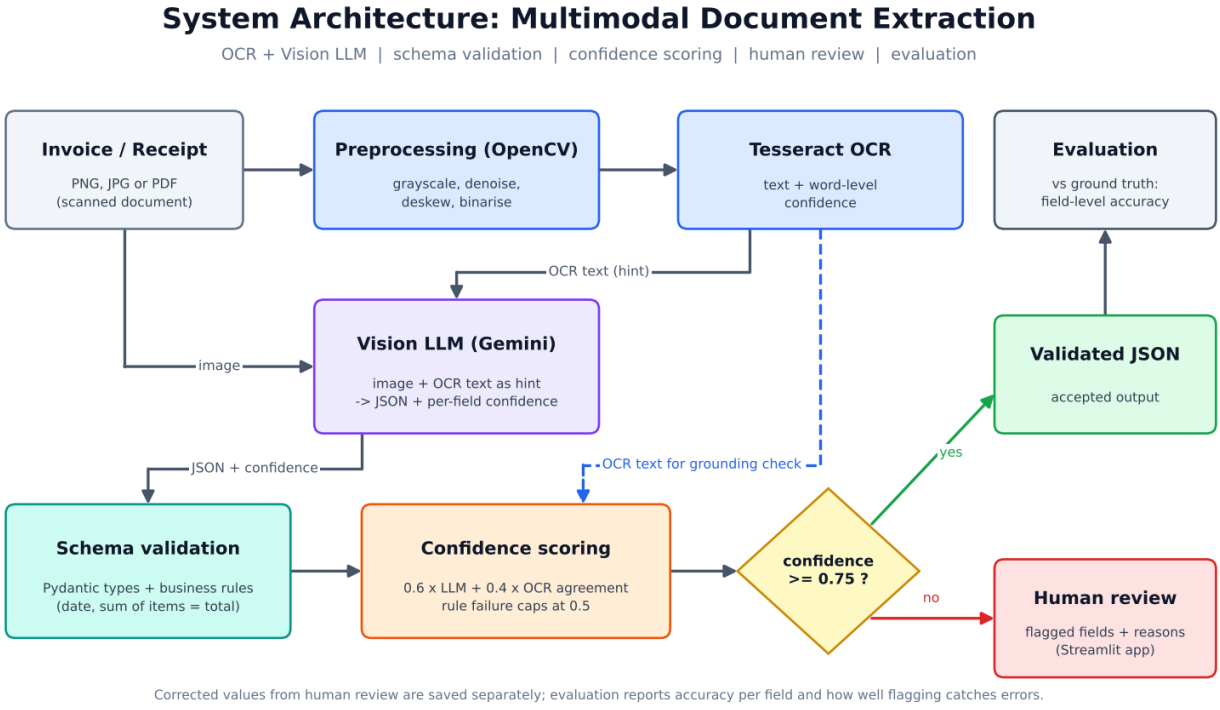


Figure 1. Architecture of the extraction pipeline.

3. Methodology

3.1 Input and preprocessing

Images (PNG, JPG, TIFF) and PDFs are supported. PDF pages are rendered to images with PyMuPDF. For OCR, OpenCV converts the page to grayscale, upscales small images, removes noise, estimates and corrects small skew angles, and binarises the image with Otsu thresholding. The vision model receives the original (resized) image.

3.2 Extraction

A single prompt instructs the model to copy values exactly as printed, to return null for missing values, to return the seller (not the client) as vendor, and to give a calibrated confidence per field. The OCR text is included as a hint, but the image is treated as the source of truth. Invalid JSON is retried.

3.3 Validation

The output is parsed into a Pydantic schema that normalises numbers (for example '$ 209,15' becomes 209.15) and dates (to ISO format). Business rules then check each field:

- invoice number present; date present, parseable and in a plausible range
- vendor present; total present and positive
- at least one line item; no missing or negative line amounts
- the sum of the line items matches the total (small tolerance)

3.4 Confidence scoring and review flags

Three independent signals are combined for every field:

Signal	Meaning
Model confidence	Confidence reported by the vision LLM for the field
OCR grounding	Is the extracted value also found in the independent OCR text?
Rule validation	Any failed business rule caps the confidence at 0.5

```
confidence = 0.6 * model_confidence + 0.4 * ocr_grounding
if a validation rule fails: confidence = min(confidence, 0.5)
if the value is missing:    confidence = 0
flag for human review if confidence < 0.75
```

Flagged fields come with a human-readable reason, and a small Streamlit app lets a reviewer correct them.

4. Evaluation setup

Data: 40 invoice images from the public katanaml invoices-donut-data-v1 dataset were downloaded together with their annotations, which were converted to the project schema. Three records with incomplete annotations (missing invoice number or total) were excluded. Documents are counted as correct per field using these rules:

Field	Matching rule
invoice_number	Equal after lower-casing and removing punctuation and spaces
invoice_date	Equal ISO date (YYYY-MM-DD)
vendor	Fuzzy partial match >= 90 (annotation also contains the address, the model returns the name)
total	Equal within 0.01
line_items	Document correct only if every row matches (amount within 0.01 and similar description) and there are no extra rows

5. Results

The evaluation covers **31 documents** with a successful prediction and complete ground truth.

Field	Correct	Accuracy
invoice_number	31/31	100.0%
invoice_date	31/31	100.0%
vendor	31/31	100.0%
line_items	31/31	100.0%
total	31/31	100.0%

- Mean field accuracy: 100.0%
- Documents with every field correct: 31/31 (100.0%)
- Line items at row level: precision 100.0%, recall 100.0%, F1 100.0%

How to read these numbers. The dataset consists of clean, synthetic invoices, and it is public, so similar documents may have been seen by the model. The vendor and description matching is deliberately lenient. The result is therefore a strong baseline on clean data, not proof of perfect behaviour on real-world scans. The flagging statistics (flagged rate per field, accuracy of values that were not flagged, share of errors caught) are listed in `reports/accuracy_report.md`.

6. Known issues and limitations

- **Free API tier.** The project was run with the free tier of the Gemini API. During the first full run the API returned HTTP 429 RESOURCE_EXHAUSTED (quota used up) and some documents failed. These documents are never counted as model errors; they are excluded and listed with the error message in the report. The run is resumable, so the remaining documents were processed after the quota allowed it.
- **Dataset.** Clean synthetic invoices; real scans (blur, stamps, handwriting, unusual layouts) will score lower.
- **Confidence is a heuristic** (model self-report, OCR agreement, rule checks), not a calibrated probability.

- **Scope.** One invoice per file; PDFs are limited to the first three pages.
- **Network and key required** for the vision model; the provider can be swapped in a single module.

7. Reproducibility

```
python download_data.py
python prepare_ground_truth.py
python run_extraction.py
python evaluate.py
streamlit run review_app.py   # optional review UI
pytest
```

The README documents setup (Python environment, Tesseract installation, API key) and the output format.

8. Future work

- Test on real, noisy scans and measure how well the flags catch the errors that appear there.
- Calibrate the confidence score (for example with isotonic regression) on a labelled validation set.
- Add a stricter vendor metric and a per-field breakdown for line-item attributes.
- Support multi-invoice PDFs and additional currencies and languages.